

Spring 2026 Topic Reviews for Math 54

Faculty: Profs. Arun Sharma and Vera Serganova
Facilitator: Jason Ferguson, jason.ferguson@berkeley.edu
Time and Location: Tu 3:30-5 on Zoom and in 201A Student Center, Th 3:30-5 on Zoom and 6 Evans
Jason's Office Hours: As of 3/16, M 12-1, TuTh 2-3 [See here](#) for the most up-to-date Drop-In Schedule



Worksheet for 3:30-5pm Tuesday, May 5, 2026's Math 54 Topic Review

Overall Topic: Fourier sine and cosine series, performing computations accurately and efficiently (B10.4, some 5.1, 5.5, 6.3, 6.4, 6.7, B10.3)

Objectives: By engaging actively with this session, participants will make progress towards being able to:

1. Given an appropriate function defined on $[0, L]$, compute its Fourier cosine or sine series
2. Given a function defined on $[0, L]$, identify its odd extension to $[-L, L]$ and even extension to $[-L, L]$ and use both to identify the functions that its Fourier sine and cosine series converge to.
3. Accurately and efficiently perform the following two key calculations from Math 54:
 - (a) Find eigenvectors of a 3×3 or larger matrix once the eigenvalue is known, including complex eigenvalues.
 - (b) Gram-Schmidt and find orthogonal projections, for both dot products and for general inner products
4. Using guidelines to identify for themselves what additional topics would benefit them to study.

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(10.3-10.4) What function does the Fourier series, Fourier sine series, or Fourier cosine series of a function converge to	
How to identify what would most help you to study next:	
Finding eigenvectors of 3x3 matrices from their eigenvalues	
Gram-Schmidt (and inner products)	

Classifying types of questions on this worksheet

I thought it would be good for your studying to divide the problems on this worksheet into the following three types:

- **R:** Recall: You need to use only one main concept or method to solve the problem, in the way it's presented in the textbook or lecture notes.
- **A:** Analyze: You need either one main concept or method to solve the problem but in a way that's different from how it's typically presented, or you need to use a group of related ideas to solve the problem.
- **S:** Synthesize: You need to combine multiple concepts that weren't originally shown to be related to solve this problem.

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(10.3-10.4) Review of Fourier series, Fourier sine series, and Fourier cosine series from 4/30's Review

Definition: (Book) Let $L > 0$ be a constant. Let f be a piecewise-continuous function on the interval $[-L, L]$. The **Fourier Series** of f is the infinite (trigonometric) series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

where:

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \text{ for } n = 0, 1, 2, \dots \quad \text{and} \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \text{ for } n = 1, 2, \dots$$

Abbreviation: “The Fourier series of $f(x)$ is $\frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(\frac{n\pi x}{L}) + b_n \sin(\frac{n\pi x}{L})]$ ” is shortened to:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right].$$

Definition: Let $L > 0$ be a constant. Let f be a piecewise-continuous function on the interval $[0, L]$. The **Fourier Cosine Series** of f is the infinite (trigonometric) series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) \quad \text{where} \quad a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \text{ for } n = 0, 1, 2, \dots$$

The **Fourier Sine Series** of f is the infinite (trigonometric) series

$$\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) \quad \text{where} \quad b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \text{ for } n = 1, 2, \dots$$

1. **A** In your own words, in one sentence:

- (Review from 4/30's Review) What does it mean to represent a function on $[-L, L]$ as a Fourier series?
- What does it mean to represent a function on $[0, L]$ as a Fourier cosine series?
- What does it mean to represent a function on $[0, L]$ as a Fourier sine series?

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(10.4) Computing Fourier Sine and Cosine Series with Tanya

Reminders: (Book) Let $L > 0$ be a constant. Let f be a piecewise-continuous function on the interval $[0, L]$.

Fourier Cosine Series on $[0, L]$: $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$ where $a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$ for $n = 0, 1, 2, \dots$.

Fourier Sine Series on $[0, L]$: $\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$ where $b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$ for $n = 1, 2, \dots$.

2. **R** Let $f(x)$ be the constant function 1. For each part, try to do as little work as you need to find an answer.
- (a) (Review from 4/30's Review) Find the Fourier series of f on $[-\pi, \pi]$.
 - (b) Find the Fourier cosine series of f on $[0, \pi]$.
 - (c) Find the Fourier sine series of f on $[0, \pi]$.

See the following graph to see both $f(x)$ on $[0, \pi]$ and partial sums of the Fourier sine series of f :

<https://www.desmos.com/calculator/xck8ge29yi>

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Extra practice for finding Fourier cosine and sine series

3. Find the Fourier cosine and sine series for the given function $f(x)$ on the given interval.

(a) **R/S** $f(x) = 2 - 5 \cos(3x)$ on $[0, \pi]$

(e) **A** $f(x) = x^2$ on $[0, L]$

(b) **S/R** $f(x) = 3 \sin(2\pi x) - 4 \sin(6\pi x)$ on $[0, 2]$

(f) **A** $f(x) = e^x$ on $[0, L]$

(c) **R** $f(x) = x$ on $[0, L]$

(g) **A** $f(x) = x(L - x)$ on $[0, L]$

(d) **R** $f(x) = x + 1$ on $[0, \pi]$

4. **A** To solve parts (a) and (b) of Problem 3 on the interval $[0, L]$ for general $L > 0$, you'd need to divide into cases based on what L is. Why?

5. **S** Solve part (a) of Problem 3 on the interval $[0, L]$ for general $L > 0$. [You're welcome to try (b), as well, but it will be even messier than (a) because there are two different trig terms.]

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(10.4) Computing Fourier, Fourier cosine, and Fourier sine series for piecewise functions

6. **A** Let $f(x)$ be the following function on $[0, 4]$:

$$f(x) = \begin{cases} x & \text{if } 0 \leq x < 2 \\ 1 & \text{if } 2 \leq x < 4 \end{cases}$$

Write down the integrals you'd need to compute to find b_n . (If you need the practice, evaluate the integrals and use them to find the Fourier sine series of f on $[0, 4]$. If you need more practice, also find the Fourier cosine series of f on $[0, 4]$.)

Answer: Sine series: $\sum_{n=1}^{\infty} \left[\frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) - \frac{2}{n\pi} \cos\left(\frac{n\pi}{2}\right) - \frac{2}{n\pi} \cos(n\pi) \right] \sin\left(\frac{n\pi x}{4}\right).$

Cosine series: $1 + \sum_{n=1}^{\infty} \left[\frac{8}{n^2\pi^2} \left(\cos\left(\frac{n\pi}{2}\right) - 1 \right) + \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) \right] \cos\left(\frac{n\pi x}{4}\right).$

The following graph shows $f(x)$ on $[0, 4]$ (green) and partial sums of the Fourier sine (purple) and cosine (orange) series of f : <https://www.desmos.com/calculator/micmll86ia>

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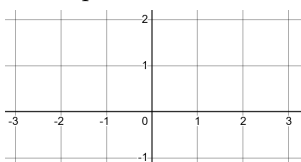
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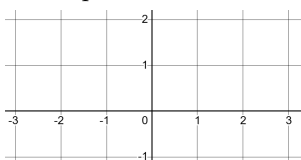
(10.4) Periodic, even, and odd extensions

7. **R** Let $f(x)$ be the function x^2 but with domain $[0, 1]$. Is f an even function?
8. **R/A** Let $f(x)$ be the function $1 - x$ with domain $(-1, 1)$. Let $g(x)$ be the function $1 - x$ with domain $(0, 1)$. Find graphs of each of the following functions for $-3 \leq x \leq 3$. (For extra practice if you need it, you can also find their formulas for $-3 \leq x \leq 3$.)

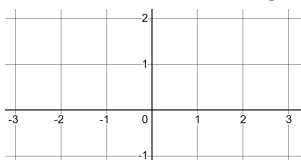
- (a) The 2-periodic extension of f .



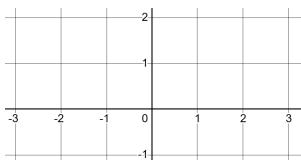
- (b) The 1-periodic extension of g .



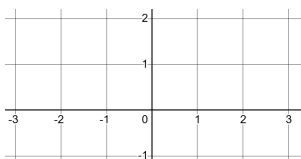
- (c) The even extension of g to $(-1, 0) \cup (0, 1)$.



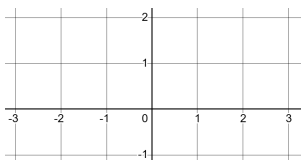
- (d) The 2-periodic extension of the even extension of g to $(-1, 0) \cup (0, 1)$.



- (e) The odd extension of g to $(-1, 0) \cup (0, 1)$.



- (f) The 2-periodic extension of the odd extension of g to $(-1, 0) \cup (0, 1)$.



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9. **R/A** Complete the following sentences.

- (a) If $f(x)$ is defined on $(0, L)$, then the Fourier sine series of f is the Fourier series of

- (b) If $f(x)$ is defined on $(0, L)$, then the Fourier cosine series of f is the Fourier series of

Suppose now g is a function defined on $(-L, L)$, and f is the same function, but considered on $(0, L)$.

- (c) If g is an even function on $(-L, L)$, then the _____ extension of f to $(-L, L)$ equals g , so the Fourier _____ series of f on $[0, L]$ is the Fourier series of g on $[-L, L]$.
- (d) If g is an odd function on $(-L, L)$, then the _____ extension of f to $(-L, L)$ equals g , so the Fourier _____ series of f on $[0, L]$ is the Fourier series of g on $[-L, L]$.

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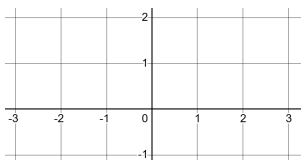
(10.3-10.4) What function does the Fourier series, Fourier sine series, or Fourier cosine series of a function converge to

10. **A** As in Exercise 9, let $f(x)$ be the function $1 - x$ with domain $(-1, 1)$. Let $g(x)$ be the function $1 - x$ with domain $(0, 1)$. Find formulas for and graphs of each of the following functions for $-3 \leq x \leq 3$.

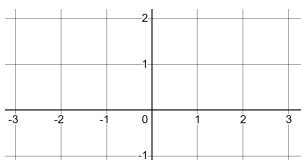
- (a) (Review from 4/30's topic review) The function that the Fourier series of f converges to.



- (b) The function that the Fourier cosine series of g converges to.



- (c) The function that the Fourier sine series of g converges to.



11. **A** Write brief summaries for how to do each of the following:

- (a) (Review from 4/30's topic review) If f is a piecewise-continuously differentiable function on $[-L, L]$, graphically find the function that the Fourier series of f converges to.

- (b) If f is a piecewise-continuously differentiable function on $(0, L)$, graphically find the function that the Fourier cosine series of f converges to.

- (c) If f is a piecewise-continuously differentiable function on $(0, L)$, graphically find the function that the Fourier sine series of f converges to.

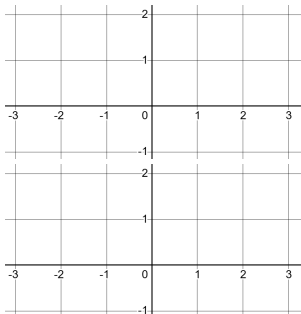
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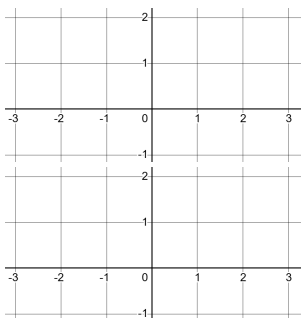


12. **A** Extra practice: Let $f(x) = \sqrt{x}$ for $0 \leq x \leq 1$. Find graphs of each of the following functions for $-3 \leq x \leq 3$.

(a) The function that the Fourier cosine series of f converges to.

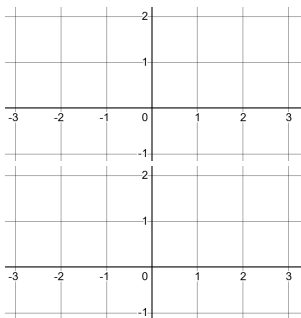


(b) The function that the Fourier sine series of f converges to.

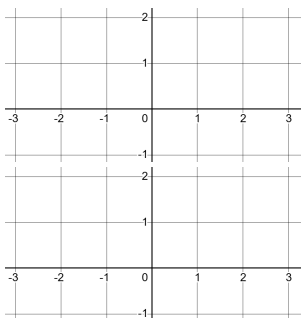


13. **A** Extra practice: Let $f(x) = \begin{cases} 0 & \text{if } 0 < x < \frac{1}{2}, \\ 1 & \text{if } \frac{1}{2} \leq x < 1. \end{cases}$. Find graphs of each of the following functions for $-3 \leq x \leq 3$.

(a) The function that the Fourier cosine series of f converges to.



(b) The function that the Fourier sine series of f converges to.



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14. **A** Let $f(x) = x + x^2$ on $[-\pi, \pi]$. It turns out that $f \sim \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left[\frac{4}{n^2}(-1)^n \cos(nx) - \frac{2}{n}(-1)^n \sin(nx) \right]$.

(You should check that if you need the practice.) By plugging in an appropriate value for x , evaluate $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

15. Extra practice:

(a) **R** Use the Fourier cosine series of $f(x) = x^2$ on $[0, \pi]$ to show that $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$ equals $\frac{\pi^2}{12}$.

(b) **A** Use the Fourier series of $f(x) = \begin{cases} 0 & -\pi \leq x < 0 \\ \pi & 0 \leq x < \pi \end{cases}$ on $-\pi < x < \pi$ to show that $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n-1} = \frac{\pi}{4}$.

How to identify what would most help you to study next:

Most Important Topics: **for Prof. Sharma's lecture**

From <https://edstem.org/us/courses/90195/discussion/8010503>

1. Row Echelon/Reduced Row Echelon Form/Elementary Row Operations - **1.1/1.2**
2. Solving Linear Systems - **1.1**
3. Linear/Matrix Transformations - **1.8/1.9**
4. Linear Independence/Dependence/Bases - **1.7/4.3**
5. Inverse Matrix - **2.2**
6. Determinants - **3.1/3.2**
7. Vector Spaces - **4.1**
8. Dimension + Rank/Nullity - **4.4/4.5**
9. Eigenvalues/Characteristic Polynomial - **5.1/5.2**
10. Regular Diagonalization - **5.3**
11. Orthogonality - **6.1/6.2/6.3**
12. Gram-Schmidt Process - **6.4**
13. Singular Value Decomposition - **7.4**
14. Solving Second Order Homogeneous Linear Equations with Constant Coefficients (Characteristic Equation) - **B4.4**
15. Solving Second Order Nonhomogeneous Linear Equations with Constant Coefficients (Undetermined Coefficients) - **B4.5**
16. Solving Systems of Linear Differential Equations - **B9.5/B9.6**

Medium Importance Topics:

1. Pivots - **1.2**
2. One-to-one/onto transformations/functions - **1.9**
3. Subspaces - **4.1**
4. Null Space/Column Space/Row Space - **4.2**

5. Similar Matrices - **5.2**
 6. Pseudo-Diagonalization of 2×2 real matrices with nonreal eigenvalues - **5.5**
 7. Orthogonal Projections - **6.3**
 8. QR Factorization - **6.4**
 9. Least Squares Curves - **6.5/6.6**
 10. Inner Product Spaces - **6.7**
 11. Symmetric Matrices/Diagonalization of Symmetric Matrices - **7.1**
 12. Superposition Principle - **B4.5**
 13. Variation of Parameters - **B4.6**
 14. Mixing Problems - **B9.5**
 15. Fourier Series (generic, sin, and cos) - **B10.3/B10.4**
-

Low Importance Topics:

1. Transformations on $\mathbb{R}^2$ (shear, dilation, etc.) - **1.9**
2. Cramer's Rule - **3.3**
3. Adjugate Matrix - **3.3**
4. Physical applications of determinant (volume etc.) - **3.3**
5. Coordinate Systems - **4.4**
6. Change of Basis - **4.6**
7. Quadratic Forms - **7.2**
8. High order differential equations as system of first order equations - **B9.1**
9. Addendum Section

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Finding eigenvectors of 3x3 matrices from their eigenvalues

16. **R** The matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ -4 & -2 & 1 \\ -1 & -1 & -1 \end{bmatrix}$ has eigenvalues -1 and $-1 \pm 2i$. Find an eigenvector for each eigenvalue.

Extra practice: Repeat: $\begin{bmatrix} -2 & 1 & -1 \\ -2 & -2 & 0 \\ -1 & 0 & -2 \end{bmatrix}$ $(-2, -2 \pm i)$; $\begin{bmatrix} 5 & -1 & -3 \\ -5 & 4 & -5 \\ -5 & 2 & -3 \end{bmatrix}$ $(2, 2 \pm i)$; $\begin{bmatrix} -7 & -15 & 15 \\ -3 & -1 & 9 \\ -4 & -7 & 10 \end{bmatrix}$ $(-2, 2 \pm 3i)$.

For even more practice, repeat with the transposes of the four matrices on this page (A and A^T will have the same eigenvalues, but not always the same eigenvectors... why?)

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Gram-Schmidt (and inner products)

17. **R** Find an orthogonal basis for $\text{Col } A$ where $A = \begin{bmatrix} -1 & 9 & 2 \\ 3 & -7 & 1 \\ 1 & -1 & 2 \\ 1 & -5 & -1 \end{bmatrix}$.

Extra practice: Repeat with one of matrices in Exercises 6.4.9-6.4.12, but with some of the columns swapped around. Find the orthogonal projection of $\begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix}$ onto $\text{Col} \begin{bmatrix} 2 & -1 \\ 1 & 3 \\ 2 & -1 \end{bmatrix}$. Find an orthogonal basis of $\text{Span}\{1, x, x^2\}$ under the inner product $\langle f, g \rangle = \int_0^1 f(x)g(x)$.